

Research Methodology in CS (CSEG3060_3)

Lecture 5: Presentation and Review Committee Assessment: preparing effective presentations, use of audio-visual aids, communication skills, handling queries, and understanding evaluation criteria of review committees

Unit 4: Technical Writing and Research Proposal Development • Mapped Course Outcome: CO5

Research Methodology in CS (CSEG3060_3)

Unit 4: Technical Writing and Research Proposal Development

Lecture 5: Presentation and Review Committee Assessment

Mapped Course Outcome: CO5 — Develop competence in presenting research proposals and defending them before review committees through effective communication and structured argumentation.

Lecture Duration: 60 Minutes

Delivery Mode: Lecture-cum-Workshop

Prerequisites: Lectures 1–4 of Unit 4

1. Learning Objectives

By the end of this lecture, the student should be able to:

1. Design and structure a research presentation that aligns with academic review standards.
 2. Select and deploy audio-visual aids in a manner that reinforces, rather than replaces, oral communication.
 3. Demonstrate effective verbal and non-verbal communication skills during a scholarly presentation.
 4. Anticipate, classify, and respond competently to questions raised by review committee members.
 5. Decode the evaluation rubrics commonly employed by review committees and align presentation strategy accordingly.
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2. Introduction

A research proposal, however scientifically rigorous, remains incomplete until it is communicated effectively to its intended audience. The culminating event in the lifecycle of a research proposal is its **defence before a Review Committee**—a panel of experts drawn from academia, industry, and (in funded projects) funding agencies. The committee's mandate is not merely to accept or reject the proposal; it is to interrogate its assumptions, methodology, novelty, feasibility, and potential impact.

This lecture addresses the **terminal competency** of the research proposal development cycle: the ability to **present, defend, and refine** a proposal under conditions of critical scrutiny. We treat presentation not as a soft skill peripheral to research, but as a **scientific act in its own right**—a form of scholarly argumentation where evidence, structure, rhetoric, and visual design converge.

The lecture is anchored in four pillars:

- **Pillar A:** Architectural design of the presentation.
- **Pillar B:** Multimodal communication through audio-visual aids.
- **Pillar C:** Interactive competence, particularly the handling of queries.
- **Pillar D:** Strategic alignment with evaluation criteria.

Each pillar is treated with reference to peer-reviewed communication research and standard practice in computer science research communities such as IEEE, ACM, and AAAI.

3. Key Concepts and Definitions

Term	Definition
Review Committee	A formally constituted panel of subject-matter experts tasked with evaluating the scientific merit, feasibility, and impact of a research proposal.
Presentation	A structured, time-bound oral exposition of a research proposal, typically supported by visual media and followed by a question-answer session.
Audio-Visual Aids (AV Aids)	Tools and media—including slides, videos, animations, whiteboards, and live demonstrations—that supplement oral communication.
Rubric	A scoring framework comprising explicit criteria, performance descriptors, and weightage used by committees to assess proposals systematically.
Elevator Pitch	A 60–90 second distillation of the research problem, motivation, and expected contribution, used at the opening of a presentation.
Back-Pocket Slide	A supplementary slide held in reserve to answer anticipated detailed queries without disrupting the flow of the main presentation.
Stage Fright Management	Cognitive and physiological techniques for controlling performance anxiety during public scholarly communication.

4. Pillar A — Designing the Effective Research Presentation

4.1 The 10/20/30 Rule (Adapted from Garr Reynolds)

A widely cited heuristic for academic presentations, originally proposed by Guy Kawasaki and refined for academic contexts:

- **10 slides** — maximum, for a 20-minute talk.
- **20 minutes** — maximum speaking time.
- **30-point font** — minimum size, ensuring legibility from the back of any standard lecture hall.

For a typical B.Tech project review of 15 minutes, the optimal structure is **8–10 slides**.

4.2 Canonical Structure for a Research Proposal Presentation

A defensible structure that mirrors the proposal document but is compressed for oral delivery:

Slide #	Content	Time Allocation	Strategic Purpose
1	Title, Author(s), Affiliation, Guide, Date	30 sec	Establishes identity and context.
2	The Elevator Pitch / Problem Statement	90 sec	Hooks the committee and frames urgency.
3	Motivation and Background	90 sec	Establishes the <i>why now</i> .
4	Literature Gap and Research Question(s)	2 min	Demonstrates awareness of state-of-the-art.
5	Objectives / Hypothesis	1 min	Crystallises the contribution claim.
6	Proposed Methodology	3–4 min	The intellectual core of the talk.
7	Expected Outcomes / Preliminary Results	2 min	Demonstrates progress and feasibility.
8	Work Plan and Timeline (Gantt Chart)	1 min	Establishes realism of execution.
9	References (selected, key 5–6)	30 sec	Signals scholarship.
10	Acknowledgements and Q&A Invitation	30 sec	Closes gracefully.

4.3 The Principle of Single-Idea-per-Slide

Each slide should communicate **one dominant cognitive unit**. The cognitive load theory of multimedia learning (Mayer, 2009) suggests that working memory can process approximately 4 ± 1 chunks of information simultaneously; overloading a slide with multiple ideas reduces comprehension.

4.4 Time Discipline

Time discipline is itself an evaluation criterion. Most committees interpret overruns as:

- Lack of rehearsal.
- Poor understanding of one's own work.
- Disrespect for the committee's schedule.

A safe strategy is to **rehearse to 80% of allotted time**, leaving 20% buffer for natural expansion under questioning.

5. Pillar B — Audio-Visual Aids

5.1 Typology of AV Aids in Research Presentations

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Audio-Visual Aids
├─ Static Visual
│   ├── Slides (PowerPoint, Beamer, Google Slides)
│   ├── Posters
│   └─ Printed handouts
├─ Dynamic Visual
│   ├── Animations and transitions
│   ├── Live system demonstration
│   ├── Video clips
│   └─ Code walk-throughs on IDE
├─ Auditory
│   ├── Spoken narration
│   ├── Audio clips / podcasts
│   └─ Music (rare, context-specific)
└─ Hybrid
    ├── Interactive simulations
    ├── Live polling tools (Mentimeter, Slido)
    └─ Augmented/Virtual Reality demos
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5.2 Design Principles for Effective Slides

Following Tufte's principles of graphical integrity and Mayer's multimedia learning principles:

5.2.1 The CRAP Framework for Slide Design

- **Contrast:** Use high contrast between text and background (dark text on light background, or vice versa).
- **Repetition:** Maintain consistent fonts, colours, and layouts across slides.
- **Alignment:** Every element should be visually connected to at least one other element.
- **Proximity:** Group related items spatially to suggest relationship.

5.2.2 Recommended Typography

- **Title font:** 28–32 pt, bold.

- **Body font:** 22–24 pt, regular.
- **Caption/footnote font:** Not less than 18 pt.
- **Recommended fonts:** Sans-serif families such as Calibri, Arial, Helvetica, or Lato—preferred over serif fonts for projected media due to better rendering at distance.

5.2.3 Colour Theory in Presentations

A 60-30-10 colour scheme is academically appropriate:

- **60% dominant colour** (typically white background or institutional colour).
- **30% secondary colour** (for headings, emphasis).
- **10% accent colour** (for highlighting novel contributions, key results).

Colour-blind-friendly palettes (e.g., viridis, ColorBrewer schemes) should be preferred when graphs and charts are presented.

5.3 Common Pitfalls in AV Aid Usage

Pitfall	Consequence	Mitigation
Reading verbatim from slides	Loss of audience engagement; committee perceives lack of depth.	Use slides as prompts, not scripts.
Excessive animation	Cognitive distraction.	Use animation only to reveal stepwise processes.
Low contrast / small fonts	Committee members at the back cannot read.	Test from the back row during rehearsal.
Cluttered graphs	Loss of insight.	Apply Tufte's data-ink ratio: maximise information, minimise chart-junk.
Embedding entire paragraphs	Information overload.	Replace prose with structured bullets and visuals.

5.4 Demonstrations: When and How

A live system demonstration can be a powerful differentiator, but it carries risk. Best practices include:

- **Pre-record the demo** as a video in case the live version fails.
- **Use a sandbox environment** with pre-loaded data.

- **Keep the demo under 90 seconds** within the main presentation; longer demos should be moved to the appendix and played only if requested.
- **Have a fallback slide** summarising the demo's outcome.

5.5 Tools and Software in Academic CS Presentations

- **Beamer (LaTeX)**: Produces publication-quality slides; ideal for theory-heavy talks.
 - **PowerPoint / Keynote**: Versatile, widely supported, suitable for design-rich content.
 - **Google Slides**: Collaborative preparation; cloud-based accessibility.
 - **Overleaf + Beamer**: Integration of manuscript and presentation visuals.
 - **Marp, RevealJS**: Markdown-based slide frameworks for technically literate presenters.
 - **OBS Studio / Loom**: For pre-recording demonstrations.
 - **Draw.io / Lucidchart**: For architecture and workflow diagrams.
 - **Matplotlib / Seaborn / Plotly**: For publication-grade figures consistent with the proposal document.
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6. Pillar C — Communication Skills

6.1 Verbal Communication

Effective verbal delivery in academic presentations comprises:

6.1.1 Pace and Pausing

- Optimal speaking rate: **130–150 words per minute** (slower than conversational speech).
- **Strategic pauses** of 1.5–2 seconds after key claims allow the committee to absorb and reflect.
- Filler words (*um, uh, like, you know*) should be minimised; silent pauses are preferable.

6.1.2 Articulation and Pronunciation

Particularly relevant for computer science terms that are often mispronounced: *LaTeX* (LAH-tek or LAY-tek), *GIF* (with soft G, contested), *Ubuntu* (oo-BOON-too), *et al.* (et al.), *ibid.* (IB-id), *viz.* (namely, *videlicet*).

6.1.3 Vocal Variation

Monotone delivery reduces retention. Use pitch variation to signal:

- **Rising pitch**: Curiosity, open-ended statements, questions.

- **Falling pitch:** Definitive claims, conclusions.
- **Lowered pitch:** Emphasis on critical claims.

6.2 Non-Verbal Communication

Albert Mehrabian's communication model, often misapplied, remains useful in emphasising that **55% of meaning in face-to-face communication** is conveyed non-verbally.

Key non-verbal channels in research presentations:

- **Eye contact:** Distribute gaze across all committee members; avoid fixating on a single member or the screen.
- **Posture:** Upright, open stance; avoid crossed arms or leaning on the podium.
- **Gestures:** Use hands to indicate transitions, enumerate points, or point to specific slide regions.
- **Movement:** Confined but purposeful; do not pace.
- **Facial expression:** Convey confidence and enthusiasm; smiles at the opening and during acknowledgements.

6.3 The Opening and the Closing

6.3.1 The Opening (First 60 Seconds)

The committee's first impression is formed rapidly and is highly durable (primacy effect). Effective openings include:

- A **provocative question**: *"What if a neural network could explain its own decisions to a radiologist in real-time?"*
- A **striking statistic**: *"Globally, 2.5 quintillion bytes of data are generated daily, yet 80% remains unstructured and unanalysed."*
- A **compelling narrative**: A brief anecdote of a real-world failure that motivates the research.
- A **clear thesis**: *"This work proposes a novel federated learning framework that reduces communication overhead by 40% while preserving differential privacy guarantees."*

6.3.2 The Closing

The closing must echo the opening and reinforce the contribution. A weak closing undermines the strongest presentation. Recommended pattern:

1. **Summary of contribution** (one sentence).

2. **Restatement of impact** (one sentence).
3. **Forward-looking statement** (one sentence).
4. **Invitation for questions** (one sentence).

Example: *"In summary, we have presented a lightweight intrusion detection system for IoT networks that achieves 96.4% accuracy with 60% lower computational cost than the state-of-the-art. This has direct implications for smart-city deployments. We will extend this to edge-fog architectures in the next phase. We welcome your questions and feedback."*

6.4 Managing Stage Fright

Stage fright is a documented physiological response mediated by the amygdala's release of adrenaline. Evidence-based techniques include:

- **Cognitive reappraisal:** Reframe the presentation as a scholarly conversation, not a performance.
 - **Systematic desensitisation:** Rehearse repeatedly in increasingly realistic conditions.
 - **Physiological regulation:** Box breathing (4-4-4-4), progressive muscle relaxation.
 - **Preparation ritual:** Arrive early, test equipment, walk the room.
 - **Power posing** (caution: research support is contested, but the psychological priming effect is real).
 - **Hydration and rest:** Avoid stimulants before the talk.
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7. Pillar D — Handling Queries from the Review Committee

7.1 Taxonomy of Committee Questions

Understanding the **type** of question is the first step to handling it competently. We propose a five-fold taxonomy:

Question Type	Intent	Example	Strategy
Clarification	Confirm understanding.	"Could you clarify what you mean by 'embedding space'?"	Paraphrase the question before answering; provide a concise definition.
Probing	Test depth of understanding.	"Why did you choose Adam optimiser over SGD for this model?"	Explain trade-offs; admit limitations if appropriate.
Challenging	Test defensibility of claims.	"Your baseline comparison is weak; have you considered X?"	Acknowledge the validity; defend or concede gracefully.
Speculative	Explore extensions.	"Could this method generalise to time-series data?"	Distinguish present scope from future work; commit if plausible.
Procedural	Clarify logistics.	"What is your data acquisition timeline?"	Answer directly; refer to the work plan slide.

7.2 The IDEAL Framework for Query Response

A mnemonic for structured response to questions:

- **I — Interpret:** Repeat or paraphrase the question to ensure accurate understanding. *"If I understand correctly, you are asking whether our approach handles concept drift..."*
- **D — Declare:** State your position clearly. *"Yes, we have addressed this through..."* or *"That is a limitation we acknowledge..."*
- **E — Evidence:** Support with data, citations, or logical reasoning.
- **A — Assess:** Evaluate the implications. *"This means our model can..."*
- **L — Link back:** Connect to the broader proposal narrative.

7.3 When You Do Not Know the Answer

Intellectual honesty is paramount. The committee is testing integrity as much as knowledge.

Recommended response:

1. **Acknowledge** clearly: *"That is an excellent point, and I do not have a definitive answer at this moment."*
2. **Demonstrate awareness:** *"However, I am aware of related work by [Author] that suggests..."*

3. Commit to follow-up: *"I will investigate this and revert with a written response within the week."*

Fabricating an answer is the single most damaging error a presenter can make.

7.4 Handling Hostile or Aggressive Questions

Occasionally, a committee member may adopt a confrontational tone. Strategies:

- **Stay calm:** Do not raise your voice or show visible irritation.
- **Validate the question:** *"I appreciate the rigour of your scrutiny."*
- **Re-route to substance:** Focus on the technical merit, not the interpersonal dynamic.
- **Appeal to data:** *"The empirical evidence suggests..."*
- **Defer to your guide** if the question lies outside your scope: *"That is a strategic decision my supervisor and I will discuss; my immediate understanding is..."*

7.5 The Back-Pocket Slide Strategy

Anticipate 5–7 likely technical questions and prepare supplementary slides containing the relevant data, equations, or diagrams. These slides are kept *after* the visible slide deck (or in a separate file) and brought up only when needed. This demonstrates preparedness without disrupting the narrative flow.

8. Understanding Evaluation Criteria of Review Committees

8.1 Generic Rubric for Research Proposal Review

Most academic and funding bodies employ a weighted rubric. The following synthesises criteria used by UGC, DST-SERB, AICTE, IEEE student grants, and similar bodies:

Criterion	Typical Weight	What Committees Look For
Problem Definition and Motivation	10–15%	Clarity of problem statement; significance to discipline/society.
Literature Review	10–15%	Coverage of seminal and recent work; identification of gaps.
Novelty and Originality	15–20%	Clear articulation of what is new; differentiation from prior work.
Methodology	20–25%	Scientific rigour; appropriateness of methods; reproducibility.
Feasibility and Work Plan	10–15%	Realistic timeline; access to resources; risk mitigation.
Expected Outcomes and Impact	10–15%	Clarity of deliverables; academic and societal value.
Presentation Quality	5–10%	Clarity, visual design, time management, Q&A handling.
Ethics and Compliance	5%	IRB/IEEE ethics, data privacy, plagiarism declarations.

8.2 Differentiating Rubrics: Academic vs. Funding Bodies

- **Academic project review (internal):** Emphasis on **methodology, learning outcomes, and demonstration**.
- **Thesis proposal defence:** Emphasis on **literature review, contribution, and methodological soundness**.
- **Funding agency review (DST, SERB, industry grants):** Emphasis on **broader impact, budget justification, deliverables, and societal benefit**.
- **Conference paper presentation:** Emphasis on **novelty, technical depth, and engagement**.

8.3 What Reviewers Write: Decoding Comments

Committee reports typically contain three classes of remarks:

- **Mandatory corrections:** Comments that *must* be addressed before approval (e.g., methodological flaw, missing ethical clearance).

- **Suggested improvements:** Comments that strengthen the proposal if addressed (e.g., additional baseline comparison).
- **Appreciative remarks:** Comments that validate the approach and signal confidence.

The first category determines acceptance; the second determines quality; the third provides strategic guidance for publication.

8.4 Strategic Alignment

Before the presentation, identify which **2–3 criteria** carry the highest weight in the given context and **design the talk to maximise scores in those areas**. For example:

- For a DST-SERB proposal, lead with **societal impact** and **deliverables**.
- For a thesis defence, lead with **methodological rigour** and **literature positioning**.
- For an industry-sponsored project, lead with **commercial viability** and **scalability**.

This strategic framing is sometimes called the "**weighted opener**" technique.

9. Practical Example: A 15-Minute CS Project Review

Consider a B.Tech final-year project on "**Privacy-Preserving Federated Learning for Healthcare IoT**". A committee will assess it on the rubric above. Below is a worked example of slide-by-slide design:

Slide 1 (Title): Project title, team of four students, supervisor Dr. X, Department of Computer Science, UPES, May 2025. Clean institutional template.

Slide 2 (Problem): *"Hospitals cannot share patient data due to HIPAA/GDPR, yet AI models require large datasets. How can we train a robust diagnostic model without centralising data?"* — Visual: split screen of two hospitals with a "no data sharing" lock icon.

Slide 3 (Background): Global healthcare data breaches 2018–2024 (graph); growth of federated learning publications (graph). Key insight in one bullet: *"FL eliminates central data aggregation, but introduces privacy, communication, and heterogeneity challenges."*

Slide 4 (Gap): Three papers summarised in three columns (FedAvg, FedProx, SCAFFOLD) with their limitations. Visual: Venn diagram showing the intersection of challenges none of them fully address.

Slide 5 (Objectives): Three numbered objectives:

1. Design a differential-privacy-aware FL framework.

2. Reduce communication rounds by 40%.
3. Validate on three healthcare benchmarks.

Slide 6 (Methodology): Architecture diagram (Clients → Secure Aggregation → Global Server with DP-Noise). Equation: $\theta_{t+1} = \theta_t - \eta \nabla L_{\text{DP}}(\theta_t; \xi_t)$ $\theta_{t+1} = \theta_t - \eta \nabla L_{\text{DP}}(\theta_t; \xi_t)$. Workflow of 5 steps.

Slide 7 (Preliminary Results): Table comparing accuracy, communication rounds, and privacy budget (ϵ) against baselines. Highlight the proposed method in a coloured cell.

Slide 8 (Work Plan): Gantt chart for the remaining 4 months: experimentation (Jan–Feb), deployment (Mar), thesis writing (Apr).

Slide 9 (References): McMahan et al. 2017, Abadi et al. 2016, Kairouz et al. 2021 — three seminal works.

Slide 10 (Closing): *"We present a privacy-aware FL framework achieving 93.2% accuracy with 42% fewer communication rounds at $\epsilon = 1.0$. We thank Dr. X, the IoT Lab, and welcome your questions."*

Anticipated back-pocket slides:

- BPS-1: Detailed convergence proof sketch.
- BPS-2: Privacy budget analysis under composition theorems.
- BPS-3: Comparison with FedBN for non-IID data.
- BPS-4: Hyperparameter sensitivity plots.

10. Common Errors to Avoid

1. **Apologising at the start:** *"Sorry, I'm nervous"* undermines confidence before the talk begins.
2. **Reading slides:** Suggests lack of depth.
3. **Hiding limitations:** Committees *will* find them; pre-empt them.
4. **Overuse of jargon:** Define technical terms on first use.
5. **Ignoring the guide's role:** Acknowledge supervisor contributions explicitly.
6. **No clear takeaway:** Every slide should advance the central thesis.
7. **Plagiarism in slides:** Reuse figures only with proper attribution; this is a serious ethical violation.
8. **No rehearsal:** Underpreparedness is the most common cause of poor reviews.
9. **Disrespecting time:** Going significantly over time forfeits goodwill.
10. **Adversarial tone during Q&A:** Treat every question as an opportunity, not an attack.

11. Summary

This lecture has examined the four pillars of effective research proposal presentation and committee assessment:

1. **Architectural design** of the presentation must mirror the proposal structure while respecting cognitive load limits and time discipline. The 10/20/30 rule and the single-idea-per-slide principle provide operational heuristics.
2. **Audio-visual aids** function as cognitive scaffolding, not as scripts. Their design must obey principles of contrast, alignment, and minimalism; their selection must match the content type (static, dynamic, interactive).
3. **Communication skills** encompass verbal, vocal, and non-verbal channels. The opening and closing of the presentation are disproportionately influential; strategic framing and graceful closure are essential.
4. **Handling queries** requires classification, structured response (the IDEAL framework), intellectual honesty, and preparedness through back-pocket slides.
5. **Evaluation criteria** are decoded through rubrics, which are context-dependent. Strategic alignment—designing the talk to maximise scores on the highest-weighted criteria—is the hallmark of an experienced presenter.

The meta-lesson is that **presentation is a scientific competence** that can be learned, rehearsed, and refined. There is no substitute for preparation, and no shortcut for integrity.

12. Exam Preparation Tips and Practice Questions

12.1 Preparation Tips

- **Internalise the rubrics** of your own institution; visit your department's review committee office to request sample rubrics from past defences.
- **Record yourself** delivering a 10-minute talk and review the video for filler words, posture, and pacing.
- **Prepare for at least 15 anticipated questions** and rehearse answers aloud.
- **Use the elevator pitch framework** to articulate your thesis in 60 seconds.
- **Conduct a mock review** with peers acting as committee members; rotate roles to understand the reviewer's perspective.

- **Review IEEE/ACM presentation guidelines** for the specific venue of your future conference presentations.

12.2 Short-Answer Questions

1. State and explain the 10/20/30 rule. In what scenario is it inapplicable?
2. Differentiate between clarifying, probing, and challenging questions. Provide one example of each from a CS research context.
3. List Mayer's six principles of multimedia learning and indicate how each applies to slide design.
4. What are the three categories of remarks in a committee report, and how should the presenter respond to each?
5. Explain the IDEAL framework for query handling with an example from your own domain.

12.3 Long-Answer / Analytical Questions

1. You are presenting a proposal on **"Deep Reinforcement Learning for Autonomous Traffic Signal Control"** to a committee comprising a control theorist, an urban planner, and an AI researcher. Design the slide structure, anticipate 7 questions (at least two from each reviewer's domain), and propose back-pocket slides for each.
2. A senior committee member challenges your proposed methodology as *"incremental and not publishable in a top-tier venue."* Outline, step-by-step, how you would respond using the IDEAL framework. What would you do if you privately agreed with the criticism?
3. Compare and contrast the evaluation rubrics of (a) a B.Tech project review committee and (b) a DST-SERB funding review committee. Identify the three criteria on which they diverge most sharply, and explain why.
4. Critically evaluate the following opening of a presentation: *"Hi everyone, my name is Rahul, and today I will present my project on machine learning. Sorry, I'm a bit nervous. So basically, my project uses CNN for image classification..."*. Rewrite it as a strong academic opening for a proposal on **"Explainable CNN Architectures for Diabetic Retinopathy Detection."**

12.4 Practical Exercise

Prepare a **10-slide deck** for the following proposal title: *"A Lightweight Blockchain-Based Framework for Secure Data Sharing in Vehicular Ad-hoc Networks (VANETs)."* The presentation must be 15 minutes long, address all canonical sections, and include annotations indicating what you would *speak* (in the speaker notes) versus what you would display (on the slide). Submit the deck along with a list of five anticipated questions and your prepared responses.

13. References and Further Reading

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End of Lecture 5 Notes

Unit 4 — Technical Writing and Research Proposal Development

Course: Research Methodology in CS (CSEG3060_3) | B.Tech (CS) | Semester 5 | UPES